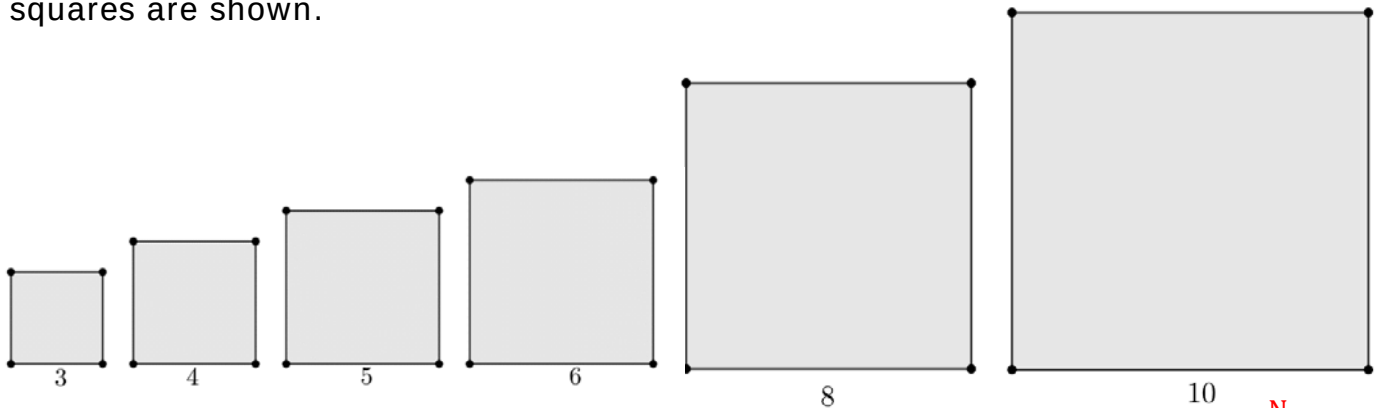


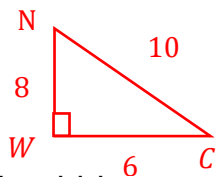
Geometry
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Six squares are shown.



The side measures of right triangle NWC are: $WC = 6$, $WN = 8$, and $NC = 10$.

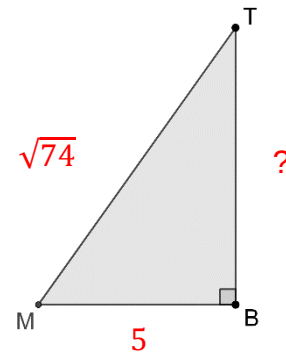


- To show a visual proof of the Pythagorean theorem which square, if any, should be placed along $\overline{WN}$?
 8×8 square
- To show a visual proof of the Pythagorean theorem which square, if any, should be placed along $\overline{WC}$?
 6×6 square

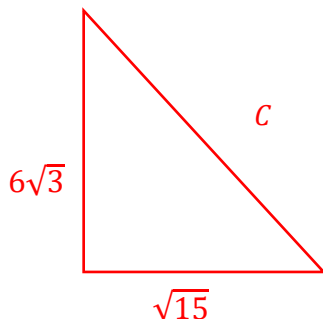
- $\triangle MTB$ is shown where $MT = \sqrt{74}$ and $MB = 5$. Find the exact length of $\overline{BT}$.

$$\begin{aligned} a^2 + b^2 &= c^2 \\ 5^2 + b^2 &= (\sqrt{74})^2 \\ 25 + b^2 &= 74 \\ \underline{-25} \quad \underline{-25} & \end{aligned}$$

$$\begin{aligned} \sqrt{b^2} &= \sqrt{49} \\ b &= 7 \\ \overline{BT} &= 7 \end{aligned}$$



- A right triangle has side lengths of $6\sqrt{3}$ and $\sqrt{15}$. Find the exact length of the hypotenuse.



$$\begin{aligned} a^2 + b^2 &= c^2 \\ (6\sqrt{3})^2 + (\sqrt{15})^2 &= c^2 \\ 108 + 15 &= c^2 \\ \sqrt{123} &= \sqrt{c^2} \\ c &= \sqrt{123} \end{aligned}$$

5. Triangle RLX is shown where $LN = NX$, $RL = 8.766$, $XR = 13.184$, and $YX = 5.39$.

Determine the length of $\overline{NY}$. If necessary, round your answer to the nearest thousandth.

1) find length of $\overline{LX}$

$$(8.766)^2 + b^2 = (13.184)^2$$

$$76.842756 + b^2 = 173.817856$$

$$b^2 = \sqrt{96.9751}$$

$$b \approx 9.848$$

$$NX = 4.924$$

2) Find $\overline{YN}$

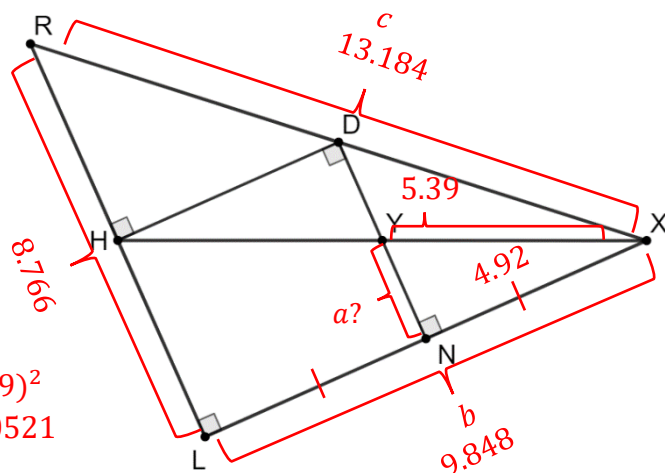
$$(4.924)^2 + a^2 = (5.39)^2$$

$$24.2458 + a^2 = 29.0521$$

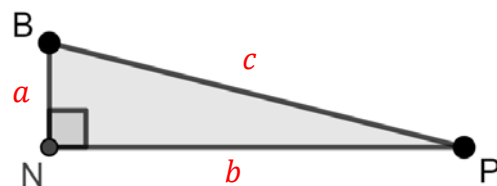
$$\sqrt{a^2} = \sqrt{4.806324}$$

$$a \approx 2.192$$

$$YN = 2.192$$



$\triangle BNP$ is shown. Complete the table by finding the missing lengths. Round your answer to the nearest thousandth. Show your work below the table. $a^2 + b^2 = c^2$



	Length of $\overline{BN}$ a leg	Length of $\overline{NP}$ b leg	Length of $\overline{BP}$ c hypotenuse
6.	48	55	73
7.	4.123	5	$\sqrt{42}$
8.	7	$\sqrt{95}$	12
9.	33	56	65
10.	5.196	$\sqrt{2}$	$\sqrt{29}$

$$\begin{aligned} 6. \quad 48^2 + b^2 &= 73^2 \\ 2304 + b^2 &= 5329 \\ \sqrt{b^2} &= \sqrt{3025} \\ b &= 55 \end{aligned}$$

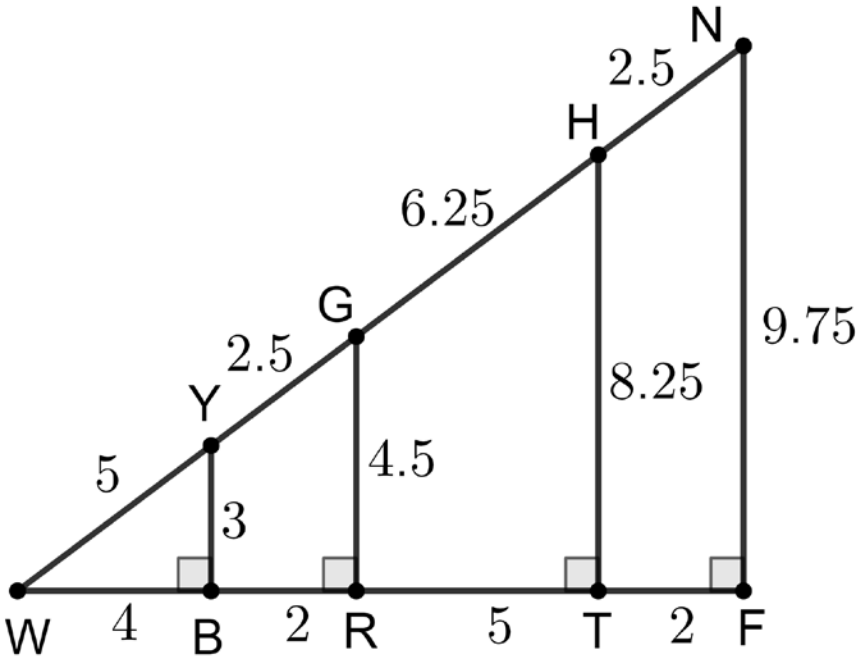
$$\begin{aligned} 7. \quad a^2 + 5^2 &= (\sqrt{42})^2 \\ a^2 + 25 &= 42 \\ \sqrt{a^2} &= \sqrt{17} \\ a &\approx 4.123 \end{aligned}$$

$$\begin{aligned} 8. \quad 7^2 + (\sqrt{95})^2 &= c^2 \\ 49 + 95 &= c^2 \\ \sqrt{c^2} &= \sqrt{144} \\ c &= 12 \end{aligned}$$

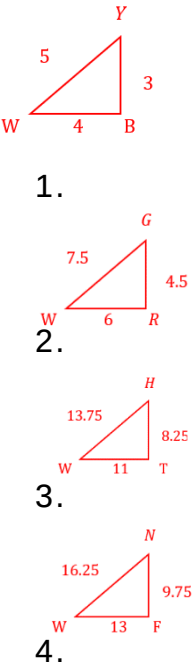
$$\begin{aligned} 9. \quad 33^2 + b^2 &= 65^2 \\ 1089 + b^2 &= 4225 \\ \sqrt{b^2} &= \sqrt{3136} \\ b &\approx 56 \end{aligned}$$

$$\begin{aligned} 10. \quad a^2 + (\sqrt{2})^2 &= (\sqrt{29})^2 \\ a^2 + 2 &= 29 \\ \sqrt{a^2} &= \sqrt{27} \\ a &\approx 5.196 \end{aligned}$$

The diagram shows measurements for $\triangle YBW$, $\triangle GRW$, $\triangle HTW$, and $\triangle NFW$.



Complete the table using $\angle W$ as the reference angle.



Triangle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
$\triangle YBW$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{3}{4}$
$\triangle GRW$	$\frac{4.5}{7.5}$	$\frac{6}{7.5}$	$\frac{4.5}{6}$
$\triangle HTW$	$\frac{8.25}{13.75}$	$\frac{11}{13.75}$	$\frac{8.25}{11}$
$\triangle NFW$	$\frac{9.75}{16.25}$	$\frac{13}{16.25}$	$\frac{9.75}{13}$

For $\triangle YBW$, $\triangle GRW$, $\triangle HTW$, and $\triangle NFW$ find the decimal equivalent for each ratio.

	Ratio	$\triangle YBW$	$\triangle GRW$	$\triangle HTW$	$\triangle NFW$
5.	$\frac{\text{opposite}}{\text{hypotenuse}}$	0.6	0.6	0.6	0.6
6.	$\frac{\text{adjacent}}{\text{hypotenuse}}$	0.8	0.8	0.8	0.8
7.	$\frac{\text{opposite}}{\text{adjacent}}$	0.75	0.75	0.75	0.75

Complete the statements.

8. The decimal equivalents of the

- ☐ $\frac{\text{opposite}}{\text{adjacent}}$
- ☐ $\frac{\text{adjacent}}{\text{hypotenuse}}$
- ☐ $\frac{\text{opposite}}{\text{hypotenuse}}$
- ☒ three

ratios for triangles YBW ,

GRW , HTW , and NFW are equal.

9. Because triangles YBW , GRW , HTW , and NFW are

- ☐ congruent
- ☒ similar

triangles,

their side lengths are

- ☐ congruent.
- ☒ proportional.

10. Since triangles YBW , GRW , HTW , and NFW share

- ☒ $\angle W$
- ☐ side $\overline{FN}$
- ☐ sides $\overline{FN}$ and $\overline{RN}$

and

are

- ☐ acute
- ☐ obtuse
- ☒ right

triangles, it is possible to define trigonometric ratios in any right

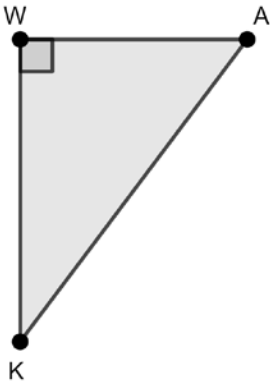
triangle.

Geometry
 Unit 7
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Triangle WKA is shown where $m\angle W = 90^\circ$.

$1. 36^2 + b^2 = 85^2$
 $1296 + b^2 = 7225$
 $\sqrt{b^2} = \sqrt{5929}$
 $b = 77$

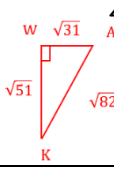
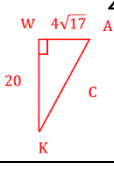
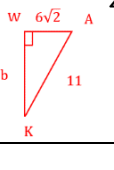


$2. 20^2 + 21^2 = c^2$
 $400 + 441 = c^2$
 $\sqrt{841} = \sqrt{c^2}$
 $c = 29$

$4. a^2 + (\sqrt{51})^2 = (\sqrt{82})^2$
 $a^2 + 51 = 82$
 $\sqrt{a^2} = \sqrt{31}$
 $a = \sqrt{31}$

Complete the table using the given information.

	$\overline{WA}$	$\overline{WK}$	$\overline{KA}$	Reference Angle	Cosine	Sine	Tangent
1.	36	77	85	$\angle A$ 	$\frac{A}{H} = \frac{36}{85}$ $\approx .424$	$\frac{O}{H} = \frac{77}{85}$ $\approx .906$	$\frac{O}{A} = \frac{77}{36}$ ≈ 2.14
2.	20	21	29	$\angle K$ 	$\frac{A}{H} = \frac{21}{29}$	$\frac{O}{H} = \frac{20}{29}$	$\frac{O}{A} = \frac{20}{21}$
3.	20	21	29	$\angle A$ 	$\frac{A}{H} = \frac{20}{29}$	$\frac{O}{H} = \frac{21}{29}$	$\frac{O}{A} = \frac{21}{20}$
4.	$\sqrt{31}$	$\sqrt{51}$	$\sqrt{82}$	$\angle K$ 	$\frac{A}{H} = \frac{\sqrt{51}}{\sqrt{82}}$	$\frac{O}{H} = \frac{\sqrt{31}}{\sqrt{82}}$	$\frac{O}{A} = \frac{\sqrt{31}}{\sqrt{51}}$

5.	$\sqrt{31}$	$\sqrt{51}$	$\sqrt{82}$		$\frac{A}{H} = \frac{\sqrt{31}}{\sqrt{82}}$	$\frac{O}{H} = \frac{\sqrt{51}}{\sqrt{82}}$	$\frac{O}{A} = \frac{\sqrt{51}}{\sqrt{31}}$
6.	$4\sqrt{17}$	20	$4\sqrt{42}$		$\frac{A}{H} = \frac{20}{4\sqrt{42}}$	$\frac{O}{H} = \frac{4\sqrt{17}}{4\sqrt{42}} = \frac{\sqrt{17}}{\sqrt{42}}$	$\frac{O}{A} = \frac{4\sqrt{17}}{20} = \frac{\sqrt{17}}{5}$
7.	$6\sqrt{2}$	7	11		$\frac{A}{H} = \frac{6\sqrt{2}}{11}$	$\frac{O}{H} = \frac{7}{11}$	$\frac{O}{A} = \frac{7}{6\sqrt{2}}$
8.	13	84	85	$\angle K$	$\frac{A}{H} = \frac{84}{85}$	$\frac{O}{H} = \frac{13}{85}$	$\frac{O}{A} = \frac{13}{84}$
9.	8	11	$\sqrt{185}$	$\angle A$	$\frac{A}{H} = \frac{8}{\sqrt{185}}$	$\frac{O}{H} = \frac{11}{\sqrt{185}}$	$\frac{O}{A} = \frac{11}{8}$
10.	9	40	41	$\angle K$	$\frac{A}{H} = \frac{40}{41}$	$\frac{O}{H} = \frac{9}{41}$	$\frac{O}{A} = \frac{9}{40}$

$$6. (4\sqrt{17})^2 + 20^2 = c^2$$

$$272 + 400 = c^2$$

$$\sqrt{672} = \sqrt{c^2}$$

$$\sqrt{16 \cdot 42} = \sqrt{c^2}$$

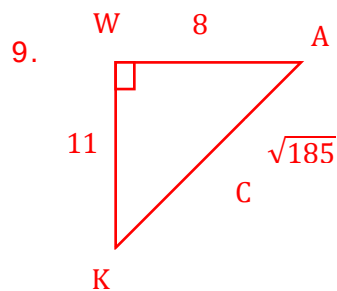
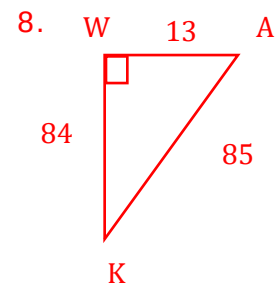
$$c = 4\sqrt{42}$$

$$7. (6\sqrt{2})^2 + b^2 = 11^2$$

$$72 + b^2 = 121$$

$$\sqrt{b^2} = \sqrt{49}$$

$$b = 7$$

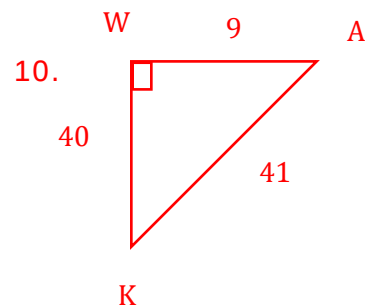


$$8^2 + 11^2 = c^2$$

$$64 + 121 = c^2$$

$$\sqrt{185} = \sqrt{c^2}$$

$$c = \sqrt{185}$$



$$9^2 + 40^2 = c^2$$

$$81 + 1600 = c^2$$

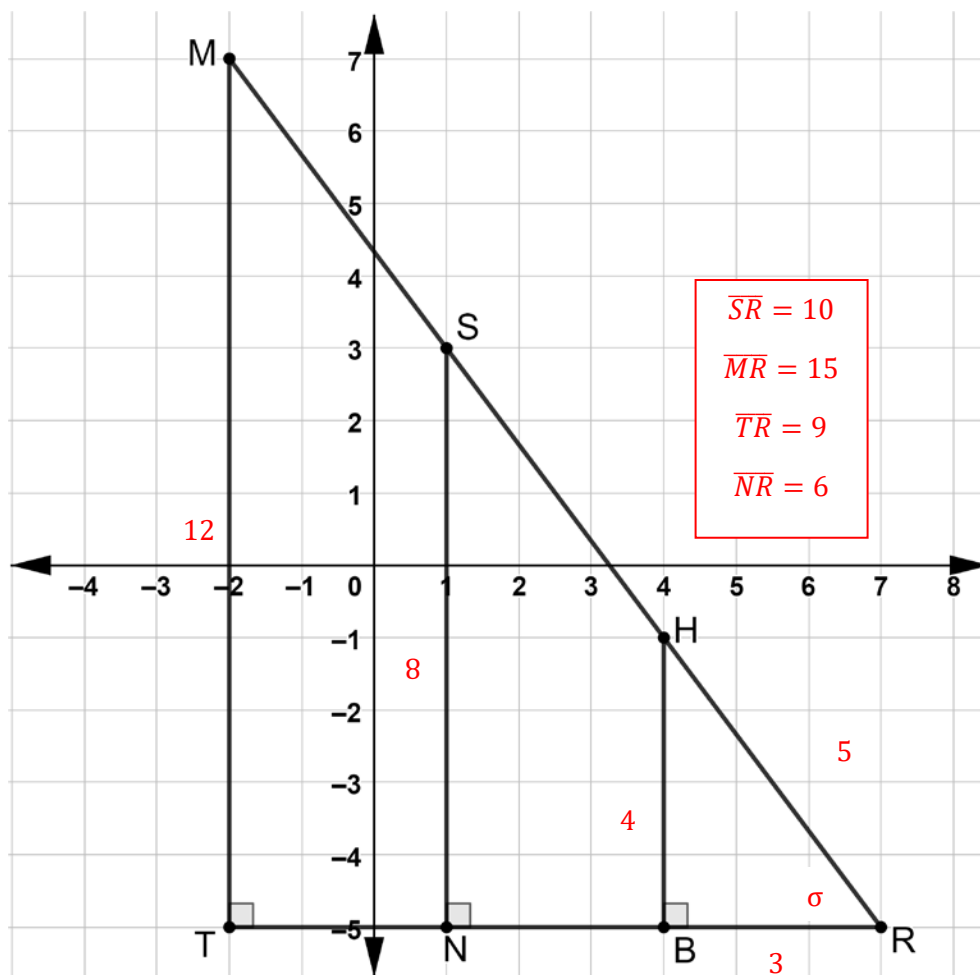
$$\sqrt{c^2} = \sqrt{1681}$$

$$c = 41$$

Geometry
Unit 7
Lesson 4 Practice

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Three right triangles are shown on the coordinate grid.



The coordinates of the vertices are shown.

$M(-2, 7)$

$S(1, 3)$

$H(4, -1)$

$R(7, -5)$

$T(-2, -5)$

$N(1, -5)$

$B(4, -5)$

1. Explain why these triangles are similar.

All three triangles share $\angle R$ and they all have a right angle.

By AA similarity, they are similar. In addition, the sides are proportional.

$$\frac{12}{9} = \frac{8}{6} = \frac{4}{3}$$

Complete the table.

	Triangle	Side	Length
2.	HRB	$\overline{BH}$	4
		$\overline{BR}$	3
		$\overline{RH}$	5 $3^2 + 4^2 = c^2$
3.	SRN	$\overline{NS}$	8
		$\overline{RN}$	6
		$\overline{SR}$	10 $6^2 + 8^2 = 10^2$
4.	MTR	$\overline{MT}$	12
		$\overline{RT}$	9
		$\overline{MR}$	15 $12^2 + 9^2 = c^2$

Using $\angle R$ as the reference angle, find the three ratios for each triangle. Write the values as a decimal rounded to the nearest thousandth.

	Triangle	Cosine	Sine	Tangent
5.	HRB	$\frac{A}{H} = \frac{3}{5} = 0.6$	$\frac{O}{H} = \frac{4}{5} = 0.8$	$\frac{O}{A} = \frac{4}{3} = 1.\bar{3}$
6.	SRN	$\frac{A}{H} = \frac{6}{10} = 0.6$	$\frac{O}{H} = \frac{8}{10} = 0.8$	$\frac{O}{A} = \frac{8}{6} = 1.\bar{3}$
7.	MTR	$\frac{A}{H} = \frac{9}{15} = 0.6$	$\frac{O}{H} = \frac{12}{15} = 0.8$	$\frac{O}{A} = \frac{12}{9} = 1.\bar{3}$

Use your calculator to find each. Round to the nearest thousandth.

	Trigonometric Ratio	Value rounded to nearest thousandth
8.	$\cos(53^\circ)$	0.602
9.	$\sin(53^\circ)$	0.799
10.	$\tan(53^\circ)$	1.327

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$\triangle PGL$ is a right triangle where $m\angle LPG = 90^\circ$, $LP = 8$, and $LG = 17$.

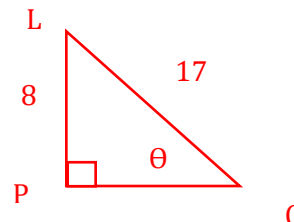
1. Determine $m\angle LGP$ using a sine ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\sin \theta = \frac{8}{17}$$

$$\theta = \sin^{-1}\left(\frac{8}{17}\right)$$

$$\theta = 28.072^\circ$$

$$m\angle LGP = 28.072^\circ$$



$\triangle BLC$ is a right triangle where $m\angle BCL = 90^\circ$, $BC = 33$, and $LC = 56$.

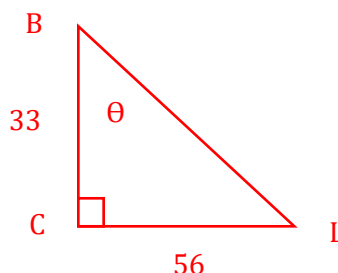
2. Determine $m\angle CBL$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\tan \theta = \frac{56}{33}$$

$$\theta = \tan^{-1}\left(\frac{56}{33}\right)$$

$$\theta \approx 59.490^\circ$$

$$m\angle CBL \approx 59.490^\circ$$



$\triangle MDZ$ is a right triangle where $m\angle DZM = 90^\circ$, $MD = 41$, and $DZ = 9$.

3. Determine the length of $\overline{ZM}$. If necessary, round your answer to the nearest thousandth.

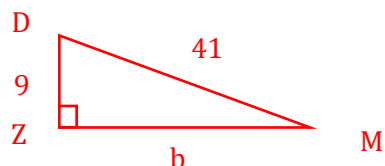
$$9^2 + b^2 = 41^2$$

$$b = \sqrt{1600}$$

$$ZM = 40$$

$$81 + b^2 = 1681$$

$$b = 40$$



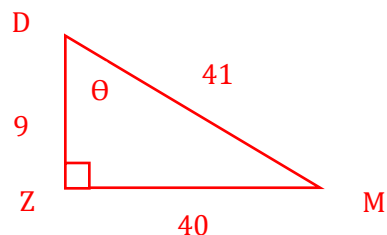
4. Determine $m\angle MDZ$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\tan \theta = \frac{40}{9}$$

$$\theta = \tan^{-1}\left(\frac{40}{9}\right)$$

$$\theta = 77.320^\circ$$

$$m\angle MDZ \approx 77.320^\circ$$



$\triangle TKB$ is a right triangle where $m\angle KTB = 90^\circ$, $KT = 92$, and $KB = 115$.

5. Determine the length of $\overline{TB}$. If necessary, round your answer to the nearest thousandth.

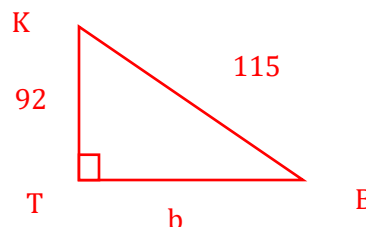
$$92^2 + b^2 = 115^2$$

$$8464 + b^2 = 13,225$$

$$b = \sqrt{4761}$$

$$b = 69$$

$$TB = 69$$

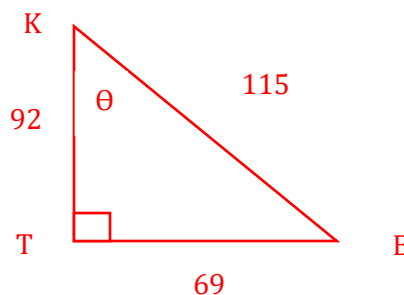


6. Determine $m\angle TKB$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\tan \theta = \frac{69}{92}$$

$$\theta = \tan^{-1} \left(\frac{69}{92} \right) \quad m\angle TKB \approx 36.870^\circ$$

$$\theta \approx 36.870^\circ$$

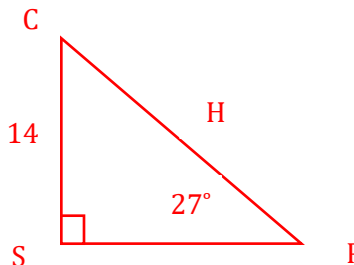


$\triangle SPC$ is a right triangle where $m\angle CSP = 90^\circ$, $m\angle CPS = 27^\circ$, and $CS = 14$.

7. Determine the length of $\overline{CP}$ using a sine ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$$\sin 27^\circ = \frac{14}{H}$$

$$H = \frac{14}{\sin 27^\circ} \approx 30.838 \quad CP \approx 30.838$$

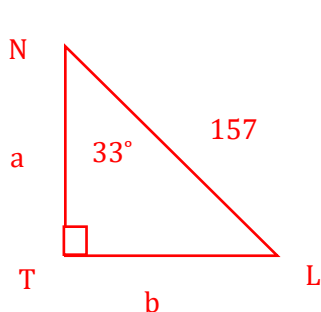


8. Explain why the tangent ratio cannot be used to determine the length of $\overline{CP}$.

Tangent is opposite over adjacent and does not include the hypotenuse. It could be used to find $\overline{SP}$ and then the Pythagorean Theorem could be used to find $\overline{CP}$, but the tangent alone cannot be used to find $\overline{CP}$ directly.

$\triangle NLT$ is a right triangle where $m\angle NTL = 90^\circ$, $m\angle TNL = 33^\circ$, and $NL = 157$.

9. Determine the length of $\overline{TL}$. Show your work. If necessary, round your answer to the nearest thousandth.



$$\sin 33^\circ = \frac{b}{157}$$

$$b = 157 \sin 33^\circ \quad TL \approx 85.508$$

$$\approx 85.508$$

10. Determine the length of $\overline{NT}$. Show your work. If necessary, round your answer to the nearest thousandth.

$$\cos 33^\circ = \frac{a}{157}$$

$$a = 157 \cos 33^\circ \quad NT \approx 131.671$$

$$\approx 131.671$$

Geometry
Unit 7
Lesson 6 Practice

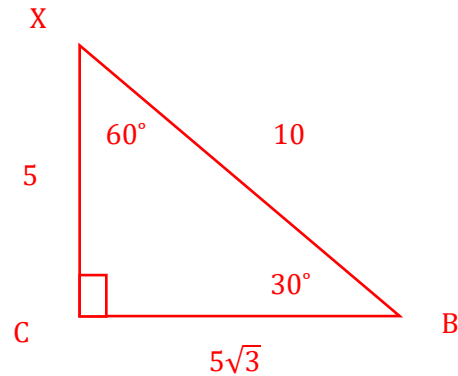
Name Answer Key
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$\triangle BXC$ is a right triangle where $m\angle XCB = 90^\circ$, $m\angle CXB = 60^\circ$, and $XC = 5$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
1.	$\overline{CB}$	$5\sqrt{3}$
2.	$\overline{XB}$	10

Hypotenuse is double the short leg (5),
so hypotenuse is 10.

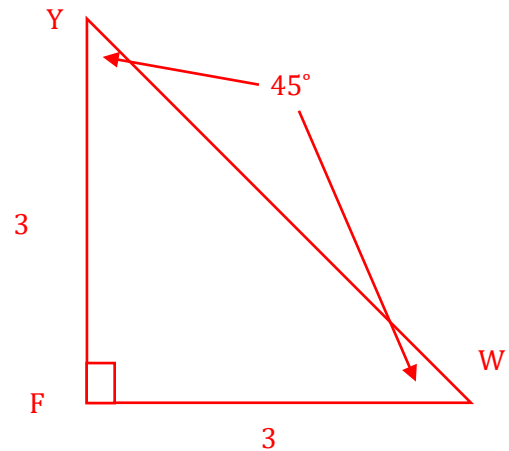
Long leg is short leg $\cdot \sqrt{3}$ ($5\sqrt{3}$).



$\triangle YFW$ is a right triangle where $m\angle YFW = 90^\circ$, $m\angle FYW = 45^\circ$, and $FY = 3$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
3.	$\overline{WF}$	3
4.	$\overline{YW}$	$3\sqrt{2}$

Isosceles so $FW = 3$, also hypotenuse
is leg $\cdot \sqrt{2}$ so $3\sqrt{2}$

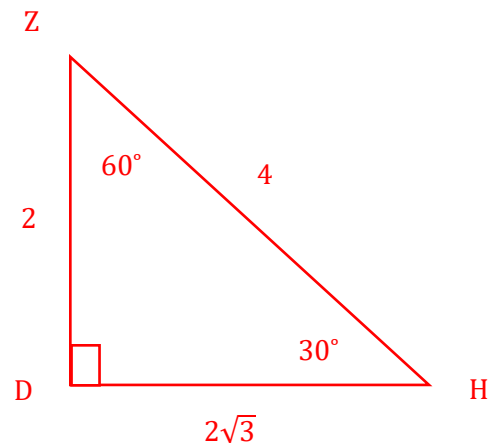


$\triangle DZH$ is a right triangle where $m\angle ZDH = 90^\circ$, $m\angle DHZ = 30^\circ$, and $ZH = 4$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
5.	$\overline{ZD}$	2
6.	$\overline{DH}$	$2\sqrt{3}$

Short leg is half the hypotenuse

Long leg is short leg $\cdot \sqrt{3}$ ($2 \cdot \sqrt{3}$)



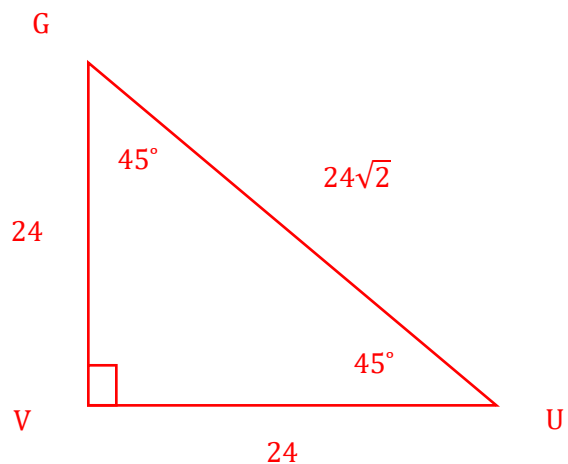
$\triangle GVU$ is a right triangle where $m\angle GVU = 90^\circ$, $m\angle UGV = 45^\circ$, and $VG = 24$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

7.

8.

Side	Length
$\overline{VU}$	24
$\overline{GU}$	$24\sqrt{2}$

$VU = 24$ since both legs are congruent
Hypotenuse is leg $\cdot \sqrt{2}$



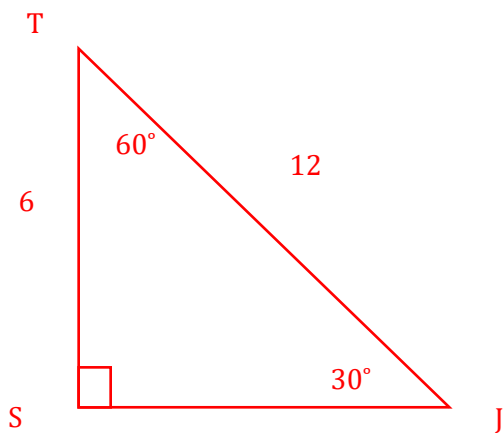
$\triangle JTS$ is a right triangle where $m\angle TSJ = 90^\circ$, $m\angle JTS = 60^\circ$, and $TJ = 12$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

9.

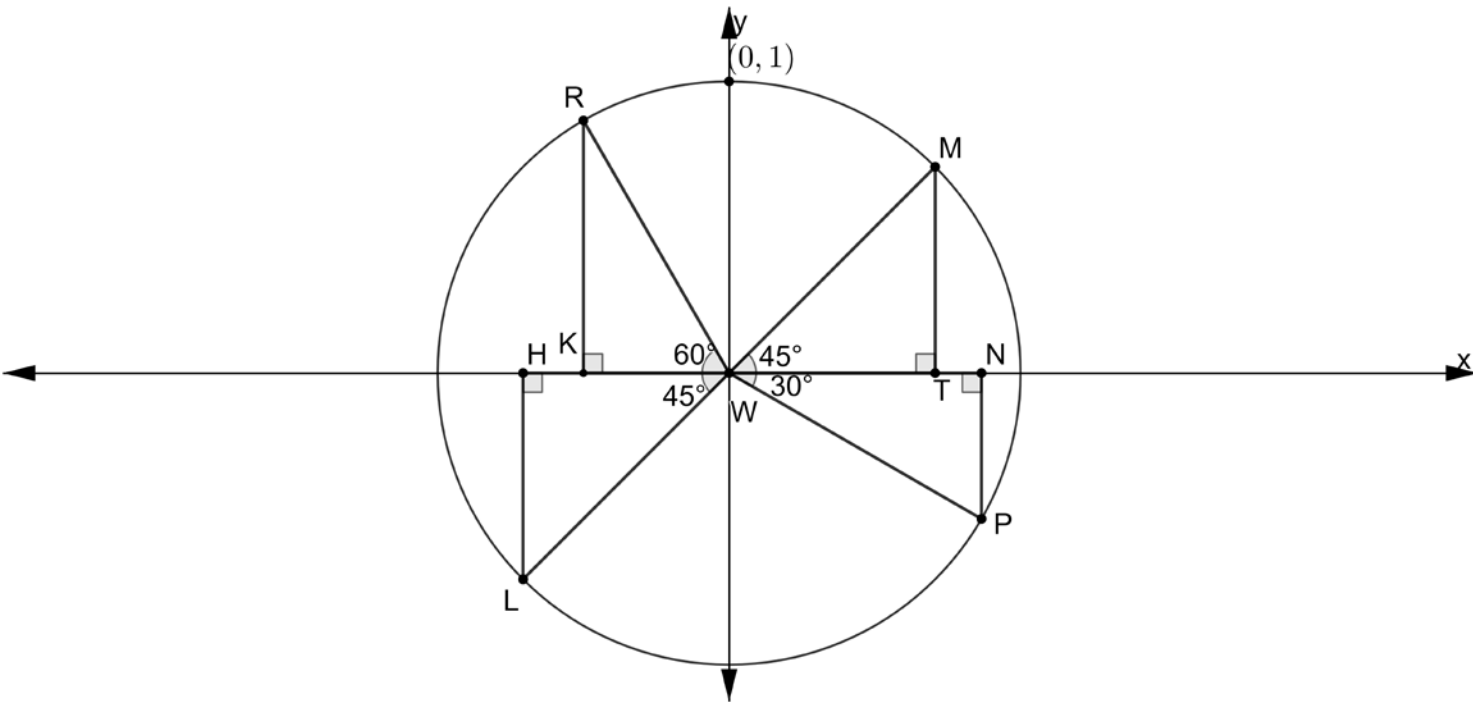
10.

Side	Length
$\overline{TS}$	6
$\overline{SJ}$	$6\sqrt{3}$

Short leg $\overline{TS}$ is half the hypotenuse.
Long leg is short leg $\cdot \sqrt{3}$ ($6\sqrt{3}$).



Circle W and right triangles WMT , WPN , RWK , and LWH are shown on the coordinate plane.



1. What is the length of the radius of circle W ? **1**
(0,0) and (0,1) are points.

Complete the table.

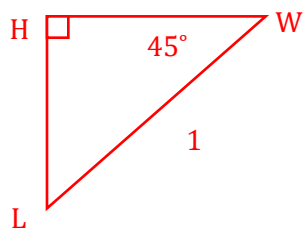
	Segment	Length
2.	$\overline{WM}$	1
3.	$\overline{PW}$	1

All radii are the same length.

Complete the table.

	Point	Coordinates
4.	L	$(-0.707, -0.707)$
5.	P	$(0.866, -0.5)$
6.	K	$(-0.5, 0)$
7.	R	$(-0.5, 0.866)$
8.	N	$(0.866, 0)$
9.	M	$(0.707, 0.707)$
10.	H	$(-0.707, 0)$

Point L :

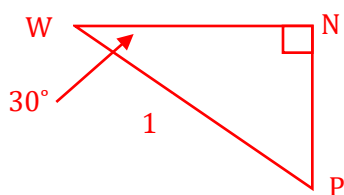


$$\sin 45 = \frac{HL}{1} \rightarrow \sin 45 = HL \approx 0.707$$

$$\cos 45 = \frac{HW}{1} \rightarrow \cos 45 = HW \approx 0.707$$

(point M has same side length of triangles)

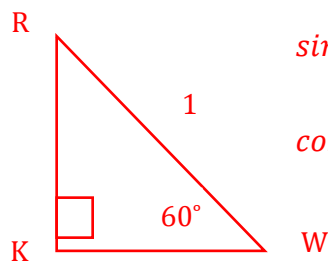
Point N and P :



$$\sin 30 = \frac{NP}{1} \rightarrow \sin 30 = NP \approx 0.5$$

$$\cos 30 = \frac{NW}{1} \rightarrow \cos 30 = NW \approx 0.8660$$

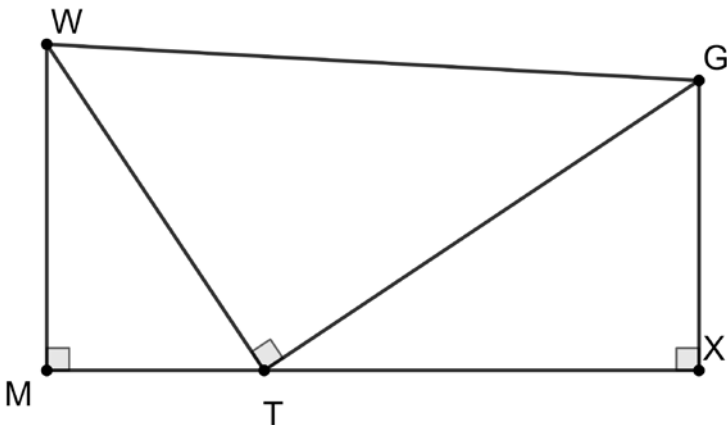
Point K and R :



$$\sin 60 = \frac{RK}{1} \rightarrow \sin 60 = RK \approx 0.866$$

$$\cos 60 = \frac{KW}{1} \rightarrow \cos 60 = KW = 0.5$$

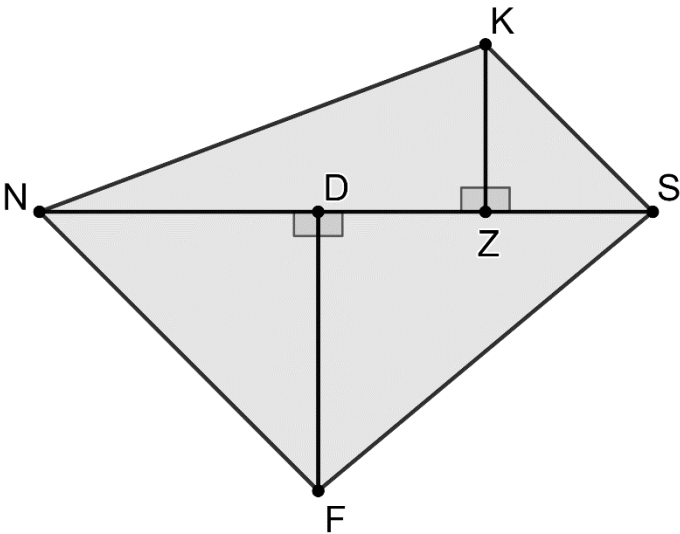
Right triangles WMT , GTW , and TXG are shown where $TX = 8$, $m\angle XTG = 34^\circ$, and $m\angle TGW = 37^\circ$.



Complete the table. In the work column, write the equation or trigonometric ratio used to determine the side measure. If necessary, round to the nearest thousandth.

	Side	Work	Side Measure
1.	$\overline{XG}$	$\tan 34 = \frac{XG}{8}$	5.396
2.	$\overline{TG}$	$\cos 34 = \frac{8}{TG}$	9.650
3.	$\overline{WT}$	$\tan 37 = \frac{WT}{9.650}$	7.272
4.	$\overline{WM}$	$\sin 56 = \frac{WM}{7.272}$	6.028
5.	$\overline{MT}$	$\cos 56 = \frac{MT}{7.272}$	4.066

Quadrilateral $NKSF$ is composed of right triangles NDF , FDS , SZK , and NZK . $ZK = 3$, $DF = 5$, $m\angle DNF = 45^\circ$, $m\angle SKZ = 45^\circ$, $m\angle FSD = 40^\circ$, and $m\angle ZNK = 21^\circ$.



Complete the table. In the work column, write the equation or trig ratio used to determine the side measure. If necessary, round to the nearest thousandth.

	Side	Work	Side Measure
6.	$\overline{KS}$	$\cos 45 = \frac{3}{KS}$	4.243
7.	$\overline{FS}$	$\sin 40 = \frac{5}{FS}$	7.779
8.	$\overline{NF}$	$\sin 45 = \frac{5}{NF}$	7.071
9.	$\overline{NK}$	$\sin 21 = \frac{3}{NK}$	8.371

10. What is the perimeter, to the nearest thousandth, of quadrilateral $NKSF$?
 $4.243 + 7.779 + 7.071 + 8.371 = 27.464$

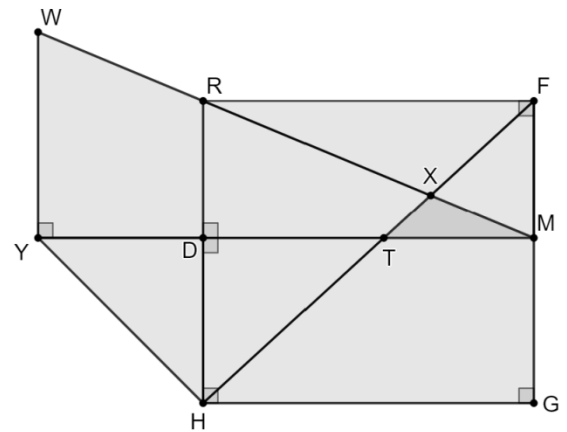
Lesson 9 Practice

Period _____

Alafia River State Park has a set of mountain bike trails. Mario draws out an area of the trails that he plans to ride. His diagram is shown.

The table shows known measurements of the trails.

$m\angle DYH = 45^\circ$	$YW = 5$ kilometers
$m\angle XMT = 23^\circ$	$DM = 8$ kilometers
$HD = 4$ kilometers	$\triangle FMR \cong \triangle DRM$
$m\angle GHF = 43^\circ$	



Complete the table. In the work column, write the equation or trigonometric ratio used to determine the side measure. Round to the nearest tenth.

	Side	Work	Side Measure
1.	$\overline{HF}$	$\cos 43 = \frac{8}{\overline{HF}}$	10.9
2.	$\overline{HY}$	$\sin 45 = \frac{4}{\overline{HY}}$	5.7
3.	$\overline{RD}$	$\tan 23 = \frac{RD}{8}$	3.4
4.	$\overline{RM}$	$\cos 23 = \frac{8}{\overline{RM}}$	8.7

5. Mario starts his ride at point W and rides along the perimeter of the park. What is the total distance, in kilometers, Mario rides? Round to the nearest tenth of a kilometer.

$$\tan 43 = \frac{GF}{8}; GF = 8 \tan 43; GF = 7.46$$

$$YD = (5.7)^2 - 4^2; YD = 4; YM = 4 + 8 = 12$$

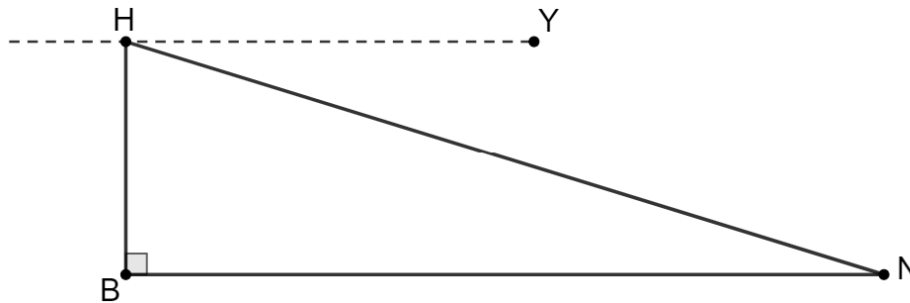
$$WM = \sqrt{5^2 + 12^2} = 13; RW = WM - RM; RW = 13 - 8.7 = 4.3$$

$$5 + 5.657 + 8 + 7.406 + 8 + 4.309 = 38.4 \text{ km}$$

6. What is the total distance, in kilometers, if Mario rides from W to Y to H to T to X to F to R ending at W . Round to the nearest tenth of a kilometer.

$$5 + 5.657 + 10.939 + 8 + 4.309 = 33.9 \text{ km}$$

Gabby is flying a drone over Lake Jackson. The drone is represented by point H and is 265 feet above Lake Jackson which is represented by point B . The angle of depression, $\angle YHN$, is the angle that Gabby will fly the drone for it to land on the fishing pier, represented by point N . The measure of $\angle YHN$ is 17° .



7. Explain why $\angle YHN$ is congruent to $\angle BNH$.

$\overline{HY}$ is parallel to the ground, $\overline{BN}$. $\overline{HN}$ is a transversal to parallel line segments $\overline{HY}$ and $\overline{BN}$, making alternate interior angles $\angle YHN$ and $\angle BNH$ congruent.

8. Determine the horizontal distance, BN , the drone is from the fishing pier. Round your answer to the nearest tenth.

$$\tan 17 = \frac{265}{BN}; BN = \frac{265}{\tan 17} \approx 866.8 \text{ ft}$$

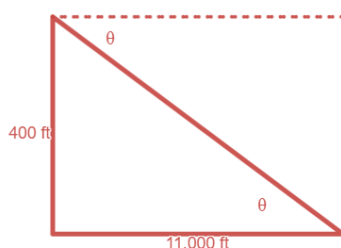
9. Determine the distance, HN , the drone must travel to reach the fishing pier. Round your answer to the nearest tenth.

$$\sin 17 = \frac{265}{HN}; HN = \frac{265}{\sin 17} \approx 906.4 \text{ ft}$$

10. Zac is also flying a drone over Lake Jackson. His drone is 11,000 feet across Lake Jackson horizontally from the fishing pier. If the height of his drone is 400 feet, what is the angle of depression Zac should fly his drone so it lands on the fishing pier? Round your answer to the nearest tenth of a degree.

$$\tan \theta = \frac{400}{11000}$$

$$\theta = \tan^{-1} \frac{400}{11000} \approx 2.082 \approx 2.1^\circ$$



Geometry
Unit 7
Lesson H1 Practice

Name Answer Key
Date _____
Period _____

In $\triangle B K X$, $m\angle K B X = 19^\circ$, $m\angle B X K = 41^\circ$, and $X K = 32$. Determine the measurement of each part of $\triangle B K X$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
1.	$m\angle X K B$	120°
2.	$\overline{X B}$	85.1
3.	$\overline{K B}$	64.5

$$\begin{aligned}
 1) \quad & m\angle X K B = 180 - (19 + 41) = 120^\circ \\
 2) \quad & \frac{\sin 19}{32} = \frac{\sin 120}{X B}; X B = 32 \frac{\sin 120}{\sin 19} = 85.121 \\
 3) \quad & \frac{\sin 19}{32} = \frac{\sin 41}{K B}; K B = 32 \frac{\sin 41}{\sin 19} = 64.484
 \end{aligned}$$

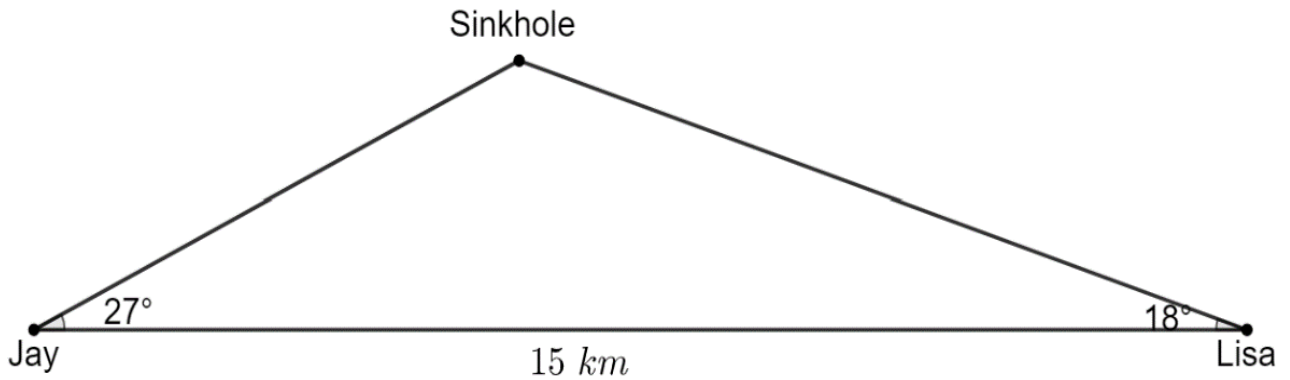
In $\triangle Z F S$, $m\angle S Z F = 47^\circ$, $Z F = 28.3$ and $S F = 51.4$. Determine the measurement of each part of $\triangle Z F S$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
4.	$m\angle F S Z$	23.7°
5.	$m\angle S F Z$	109.3°
6.	$\overline{S Z}$	66.4°

$$\begin{aligned}
 4) \quad & \frac{\sin 47}{51.4} = \frac{\sin \angle F S Z}{28.3}; \sin \angle F S Z = \frac{28.3 \sin 47}{51.4}; \angle F S Z = 23.7^\circ \\
 5) \quad & m\angle S F Z = 180 - (47 + 23.475) = 109.255 \\
 6) \quad & \frac{\sin 47}{51.4} = \frac{\sin 109.255}{S Z}; S Z = \frac{51.4 \cdot \sin 109.255}{\sin 47}; S Z = 66.349
 \end{aligned}$$

Jay and Lisa are both hiking in the Leon Sinks Geological Area. They are currently 15 kilometers (km) apart. They plan to meet up at one of the sinkholes in the park.

7. How far, to the nearest hundredth of a kilometer, does Jay hike?



Sinkhole angle: $180 - (27 + 18) = 135^\circ$

Jay to Sinkhole: $\frac{\sin 18}{x} = \frac{\sin 135}{15}$; $x \approx 6.56$ km

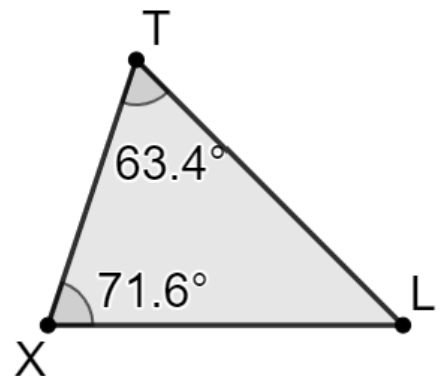
8. How far, to the nearest hundredth of a kilometer, does Lisa hike?

$\frac{\sin 27}{y} = \frac{\sin 135}{15}$; $y \approx 9.63$ km

Shanice runs in a competitive obstacle course. The obstacle course and some of its measurements are shown. The distance between point T and point X is 3.2 miles.

9. Write an equation that can be used to determine the distance, in kilometers, between point T and point L .

$\frac{\sin 71.6}{x} = \frac{\sin(180-63.4-71.6)}{5}$; (3.2 miles = 5 km)



10. Determine the perimeter, in miles, of the obstacle course.

$TL: \frac{\sin 71.6}{x} = \frac{\sin(180-63.4-71.6)}{3.2}$; $TL \approx 4.294$

$XL: \frac{\sin 45}{3.2} = \frac{\sin 63.4}{y}$; $XL \approx 4.046$

Perimeter: $3.2 + 4.294 + 4.046 = 11.54$ miles

Geometry
Unit 7
Lesson H2 Practice

Name Answer Key
Date _____
Period _____

In $\triangle GHS$, $m\angle HGS = 22^\circ$, $HG = 27$, and $SG = 14.9$. Determine the measurement of each part of $\triangle GHS$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
1.	$\angle SHG$	23°
2.	$\angle HSG$	135°
3.	$\overline{HS}$	14.3

$$1. \quad 14.9^2 = 27^2 + 14.32^2 - 2(27)(14.32) \cos B$$

$$773.28 \cos B = 712.0524$$

$$B = 22.954^\circ$$

$$2. \quad 27^2 = 14.9^2 + 14.32^2 - 2(14.9)(14.32) \cos a$$

$$426.736 \cos a = -301.928$$

$$a = 135.03$$

$$3. \quad x^2 = 27^2 + 14.9^2 - 2(27)(14.9) \cos 22 = 729 + 222.01 - 746.012 = 204.998$$

$$x = 14.32$$

In $\triangle PXN$, $PX = 17$, $PN = 28$, and $XN = 30$. Determine the measurement of each part of $\triangle PXN$. Show your work below the table. If necessary, round to the nearest tenth.

	Part	Measurement
4.	$\angle PXN$	66.6°
5.	$\angle PNX$	33.9°
6.	$\angle XPN$	79.5°

$$4. \quad 28^2 = 30^2 + 17^2 - 2(30)(17) \cos x$$

$$1020 \cos x = 405$$

$$x = 66.6$$

$$5. \quad 17^2 = 30^2 + 28^2 - 2(30)(28) \cos n$$

$$1680 \cos n = 1395$$

$$n = 33.9$$

$$6. \quad 30^2 = 17^2 + 28^2 - 2(17)(28) \cos p$$

$$952 \cos p = 173$$

$$p = 79.5$$

Austyn is checking a water trough in her pasture before walking to the barn. She enters the pasture through a gate near her house. The distance from the gate, point G , to the water trough, point W , is 52 yards. The distance from the gate to the barn, point N , is 37.5 yards. The distance from the water trough to the barn is 31 yards. Determine each angle measure.

7.
8.
9.

Angle	Measurement
$\angle NGW$	36°
$\angle GWN$	46°
$\angle WNG$	98°

10.
- In $\triangle DTF$, $DF = 10$, $TD = 42$, and $TF = 46$. Complete the statement about $\triangle DTF$.

$\triangle DTF$ is a

o right

o acute

o obtuse

o right

triangle because

o one of the angles is

o all of the angles are

o equal to 90° .

o less than 90° .

o greater than 90° .

$\overline{FT}$ is longest segment so $\angle FDT$ is largest angle.

$$46^2 = 10^2 + 42^2 - 2(10)(42) \cos b$$

$$\cos b = \frac{-252}{840}$$

$$b = 107.46^\circ$$

Geometry
Unit 7
Lesson H3 Practice

Name Answer Key
Date _____
Period _____

Determine the area of each triangle using the given information. Round to the nearest tenth.

	Triangle	Given Information			Area
1.	$\triangle BHZ$	$HB = 7$	$ZB = 12$	$m\angle HBZ = 64^\circ$	$= \frac{1}{2}(7)(12)(\sin 64)$ $= 37.7 \text{ units}^2$
2.	$\triangle YCJ$	$CY = 100$	$JY = 63$	$m\angle JYC = 80^\circ$	$= \frac{100 \cdot 63 \cdot \sin 80}{2}$ $= 3,102.1 \text{ units}^2$
3.	$\triangle KDN$	$DN = 7.2$	$NK = 6.3$	$m\angle DNK = 95^\circ$	$= \frac{1}{2}(7.2)(6.3) \sin 95$ $= 22.6 \text{ units}^2$
4.	$\triangle FSG$	$SF = 3\sqrt{5}$	$GS = 2\sqrt{7}$	$m\angle FSG = 36^\circ$	$= \frac{1}{2}(3\sqrt{5})(2\sqrt{7})(\sin 36)$ $= 10.4 \text{ units}^2$

5.	$\triangle WQL$	$QW = 8.4$	$LW = 9.9$	$m\angle QWL = 68^\circ$	$= \frac{1}{2}(8.4)(9.9)(\sin 68)$ $= 38.6 \text{ units}^2$
6.	$\triangle BRS$	$SR = \frac{78}{5}$	$RB = \frac{119}{8}$	$m\angle SRB = 42^\circ$	$= \frac{1}{2}\left(\frac{78}{5}\right)\left(\frac{119}{8}\right)(\sin 42)$ $= 77.6 \text{ units}^2$
7.	$\triangle TXD$	$TX = 2.6$	$TD = 5.1$	$m\angle XTD = 16^\circ$	$= \frac{1}{2}(2.6)(5.1) \sin 16$ $= 1.8 \text{ units}^2$
8.	$\triangle YNK$	$NY = \sqrt{11}$	$NK = \sqrt{11}$	$m\angle KNY = 72^\circ$	$= \frac{1}{2}(\sqrt{11})(\sqrt{11}) \sin 72$ $= 5.2 \text{ units}^2$
9.	$\triangle LGR$	$GL = 521$	$LR = 94$	$m\angle RLG = 179^\circ$	$= \frac{1}{2}(521)(94) \sin 179$ $= 427.4 \text{ units}^2$
10.	$\triangle ABC$	$BC = 5\sqrt{3}$	$AB = 4\sqrt{7}$	$m\angle ABC = 89^\circ$	$= \frac{1}{2}(5\sqrt{3})(4\sqrt{7}) \sin 89$ $= 45.8 \text{ units}^2$